

UML DIAGRAM

SUBMITTED BY;

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PROJECT: DONATION MANAGEMENT SYSTEM

1. UML DIAGRAMS

1.1 USE CASES

1.1.1 USE CASE 1: REGISTER NEW ACCOUNT

- **Use Case ID:** UC01
- **Use Case Name:** User Registration
- **Description:** A new user registers as a Donor or Receiver to access the system.
- **Primary Actor:** User
- **Secondary Actor:** None

Preconditions:

- User is not already registered.

Postconditions:

- A new user account is created successfully.

Main Flow:

1. User opens the registration page
2. User enters name, email/phone number, role, and password.
3. User submits the form.
4. System validates the details.
5. System creates the account.
6. User is redirected to login.

Alternative Flow:

1. If the email already exists or the input is invalid, an error is displayed.
2. User corrects the input and resubmits.

1.1.2 USE CASE 2: USER LOGIN

- **Use Case ID:** UC02
- **Use Case Name:** User Login
- **Description:** Registered users log in using secure authentication (JWT).
- **Primary Actor:** User (Donor / Receiver / Admin)
- **Secondary Actor:** None

Preconditions :

- User must be registered.

Postconditions:

- User is authenticated and logged in.

Main Flow:

1. User navigates to login page.
2. User enters credentials.
3. System verifies credentials.
4. System issues authentication token.
5. User is redirected to dashboard.

Alternative Flow:

1. If credentials are invalid, error is shown.
2. User retries login.

1.1.3 USE CASE 3: CREATE ITEM REQUEST

- **Use Case ID:** UC03

- **Use Case Name:** Create Item Request
- **Description:** Receiver submits a request for required items.
- **Primary Actor:** Receiver
- **Secondary Actor:** None

Preconditions:

- Receiver must be logged in.

Postconditions:

- Item request is created with status *Pending*.

Main Flow:

1. Receiver navigates to request page.
2. Receiver enters item details (name, category, condition, quantity, usage duration).
3. Receiver submits the request.
4. System validates the request logic.
5. System saves the request
6. Confirmation messages are displayed.

Alternative Flow:

1. If data is invalid, the system shows error.
2. Receiver corrects and resubmits.

1.1.4 USE CASE 4: VIEW OWN ITEM REQUESTS

- **Use Case ID:** UC04
- **Use Case Name:** View Request History
- **Description:** Receiver views only their own item requests.
- **Primary Actor:** Receiver
- **Secondary Actor:** None

Preconditions :

- Receiver must be logged in.

Postconditions:

- Receiver's request list is displayed.

Main Flow:

1. Receiver opens "My Requests".
2. System filters requests by logged-in receiver.
3. System displays request details and status.

1.1.5 USE CASE 5: CREATE DONATION

- **Use Case ID:** UC05
- **Use Case Name:** Create Donation
- **Description:** Donor lists items for donation with images.
- **Primary Actor:** Donor
- **Secondary Actor:** Cloud Storage Service

Preconditions:

- Donor must be logged in.

Postconditions:

- Donation is created successfully.

Main Flow:

1. Donor navigates to donation creation page.
2. Donor enters item details.
3. Donor uploads item images.

4. System uploads images to cloud storage.
5. System saves donation data.
6. Confirmation is shown.

Alternative Flow:

1. If image upload fails, error is displayed.
2. Donor retries upload.

4.1.6 USE CASE 6: VIEW OWN DONATIONS

- **Use Case ID:** UC06
- **Use Case Name:** View Donations
- **Description:** Donor views donations created by them.
- **Primary Actor:** Donor
- **Secondary Actor:** None

Preconditions:

- Donor must be logged in.

Postconditions:

- Donor's donation list is displayed.

Main Flow:

1. Donor opens "My Donations".
2. System filters donations by logged-in donor.
3. System displays donation details.

4.1.7 USE CASE 7: UPDATE DONATION

- **Use Case ID:** UC07
- **Use Case Name:** Update Donation
- **Description:** Donor updates donation details before assignment.
- **Primary Actor:** Donor
- **Secondary Actor:** None

Preconditions:

- Donation exists and is not fulfilled.

Postconditions:

- Donation details are updated.

Main Flow:

1. Donor selects a donation.
2. Donor edits donation details.
3. System validates changes.
4. System saves updates.

4.1.8 USE CASE 8: VIEW ALL ITEM REQUESTS

- **Use Case ID:** UC08
- **Use Case Name:** View All Requests
- **Description:** Admin views all receiver item requests.
- **Primary Actor:** Admin
- **Secondary Actor:** None

Preconditions:

- Admin must be logged in.

Preconditions:

- All item requests are displayed.

Main Flow:

1. Admin opens request management page.
2. System retrieves all requests.
3. System displays request list.

4.1.9 USE CASE 9: APPROVE OR REJECT REQUEST

- **Use Case ID:** UC09
- **Use Case Name:** Manage Item Request
- **Description:** Admin approves or rejects item requests.
- **Primary Actor:** Admin
- **Secondary Actor:** None

Preconditions:

- Request status is *Pending*.

Postconditions:

- Request status updated to *Approved* or *Rejected*.

Main Flow:

1. Admin selects a pending request.
2. Admin approves or rejects it.

3. System updates request status.
4. Receiver is notified.

4.1.10 USE CASE 10: MATCH DONATION WITH REQUEST

- **Use Case ID:** UC10
- **Use Case Name:** Match Donation
- **Description:** Admin assigns a suitable donation to an approved request.
- **Primary Actor:** Admin
- **Secondary Actor:** None

Preconditions:

- Request is approved.
- Donation is available.

Postconditions:

- Donation order is created.

Main Flow:

1. Admin selects an approved request.
2. Admin selects a matching donation.
3. System creates donation order.
4. System updates statuses.

4.1.11 USE CASE 11: UPDATE REQUEST STATUS

- **Use Case ID:** UC11
- **Use Case Name:** Update Request Status
- **Description:** Admin marks a request as fulfilled after completion.
- **Primary Actor:** Admin
- **Secondary Actor:** None

Preconditions:

- Donation has been assigned.

Postconditions:

- Request status becomes *Fulfilled*.

Main Flow:

1. Admin selects assigned request.
2. Admin updates status to fulfilled.
3. System saves changes.

4.1.12 USE CASE 12: LOGOUT

- **Use Case ID:** UC12
- **Use Case Name:** Logout
- **Description:** User logs out of the system securely.
- **Primary Actor:** User

- **Secondary Actor:** None

Preconditions:

- User must be logged in.

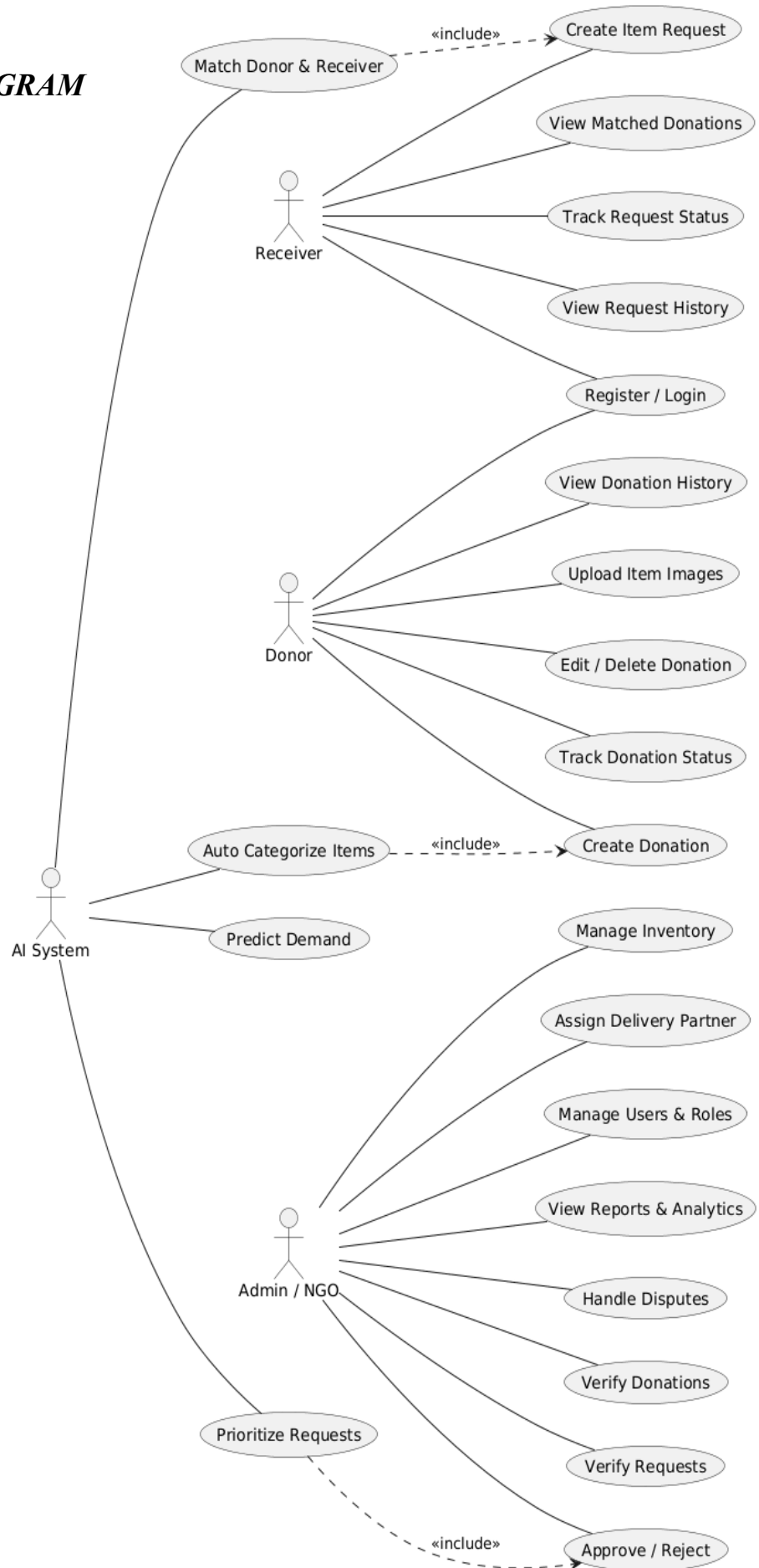
Postconditions:

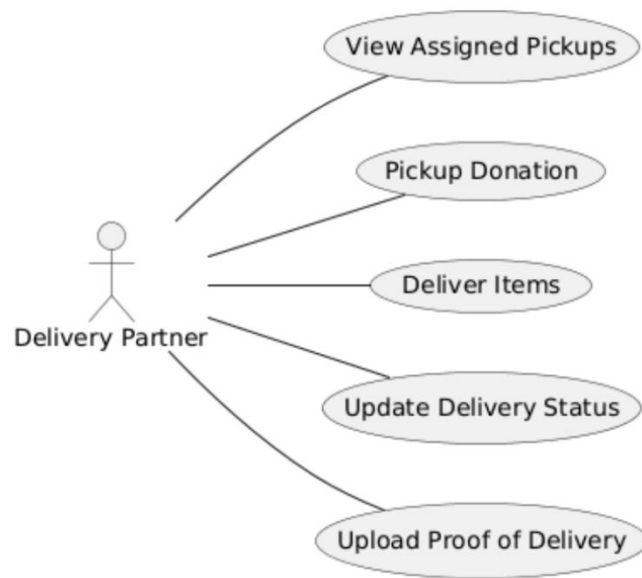
- User session is terminated.

Main Flow:

1. User clicks logout.
2. System invalidates the authentication token.
3. User is redirected to the login page.

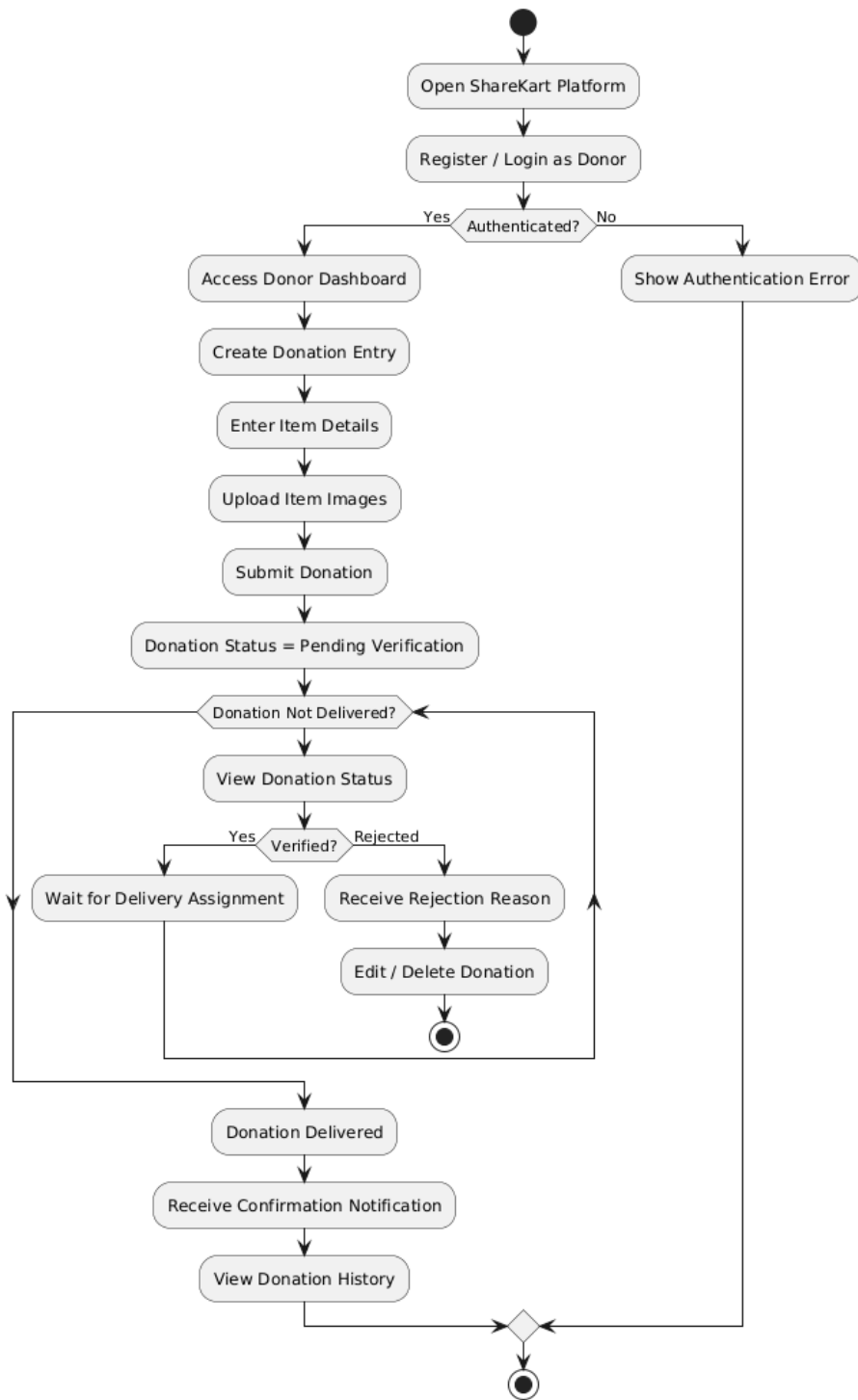
1.2 USE CASE DIAGRAM



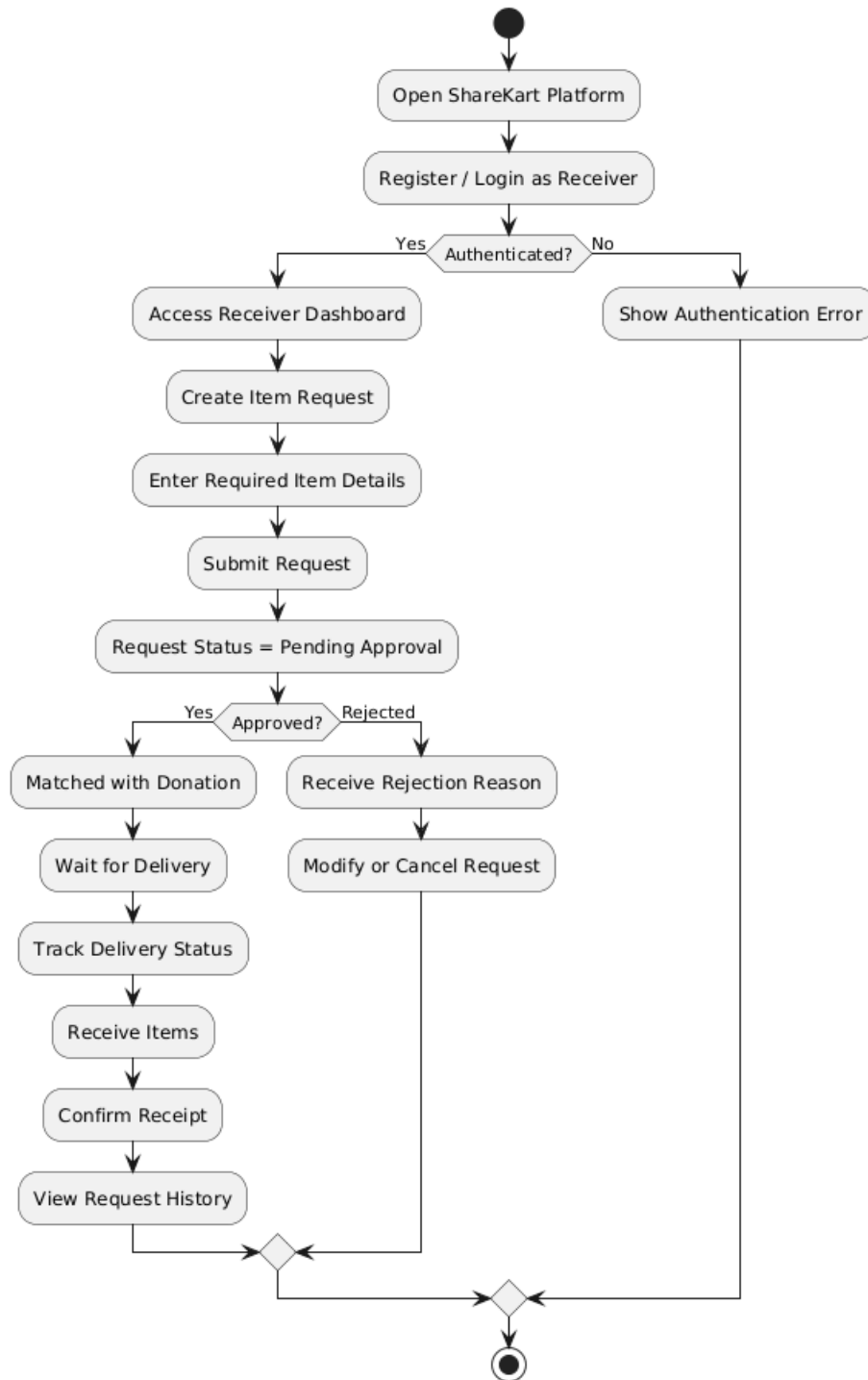


1.3 ACTIVITY DIAGRAM

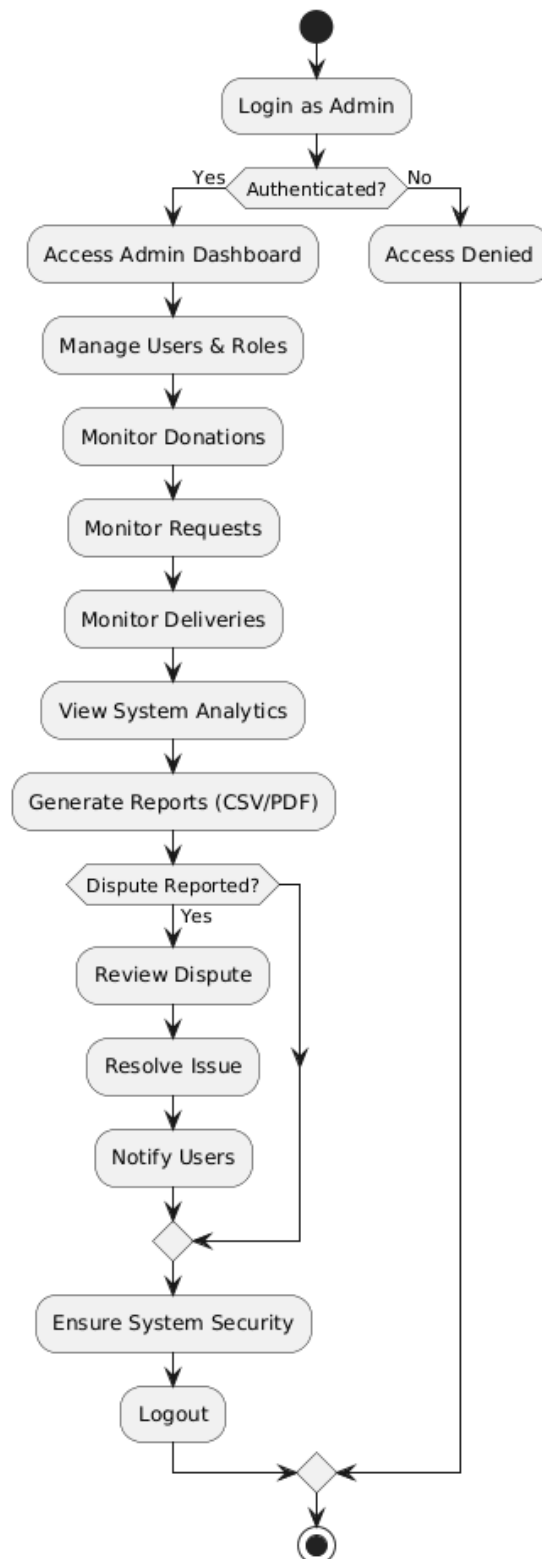
1.3.1 DONOR



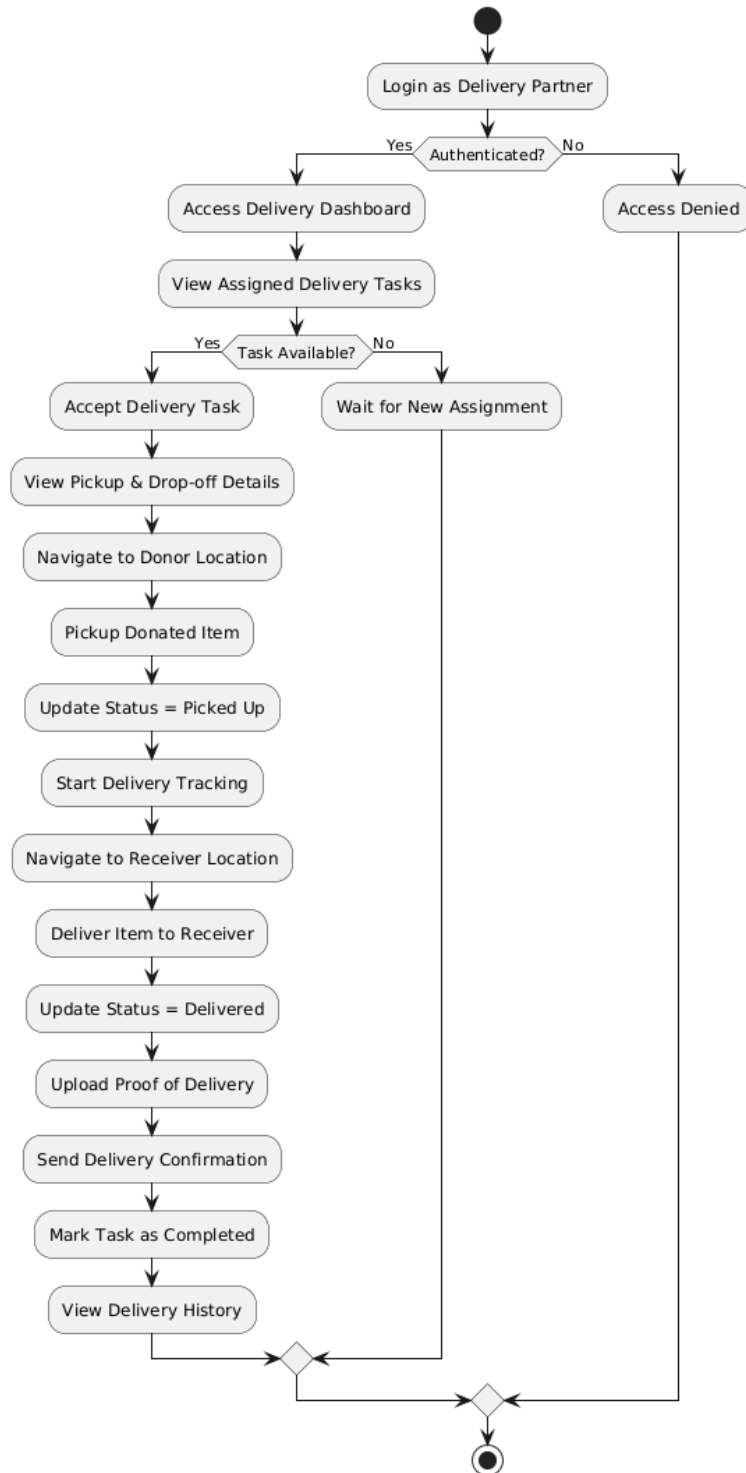
1.3.2 RECEIVER



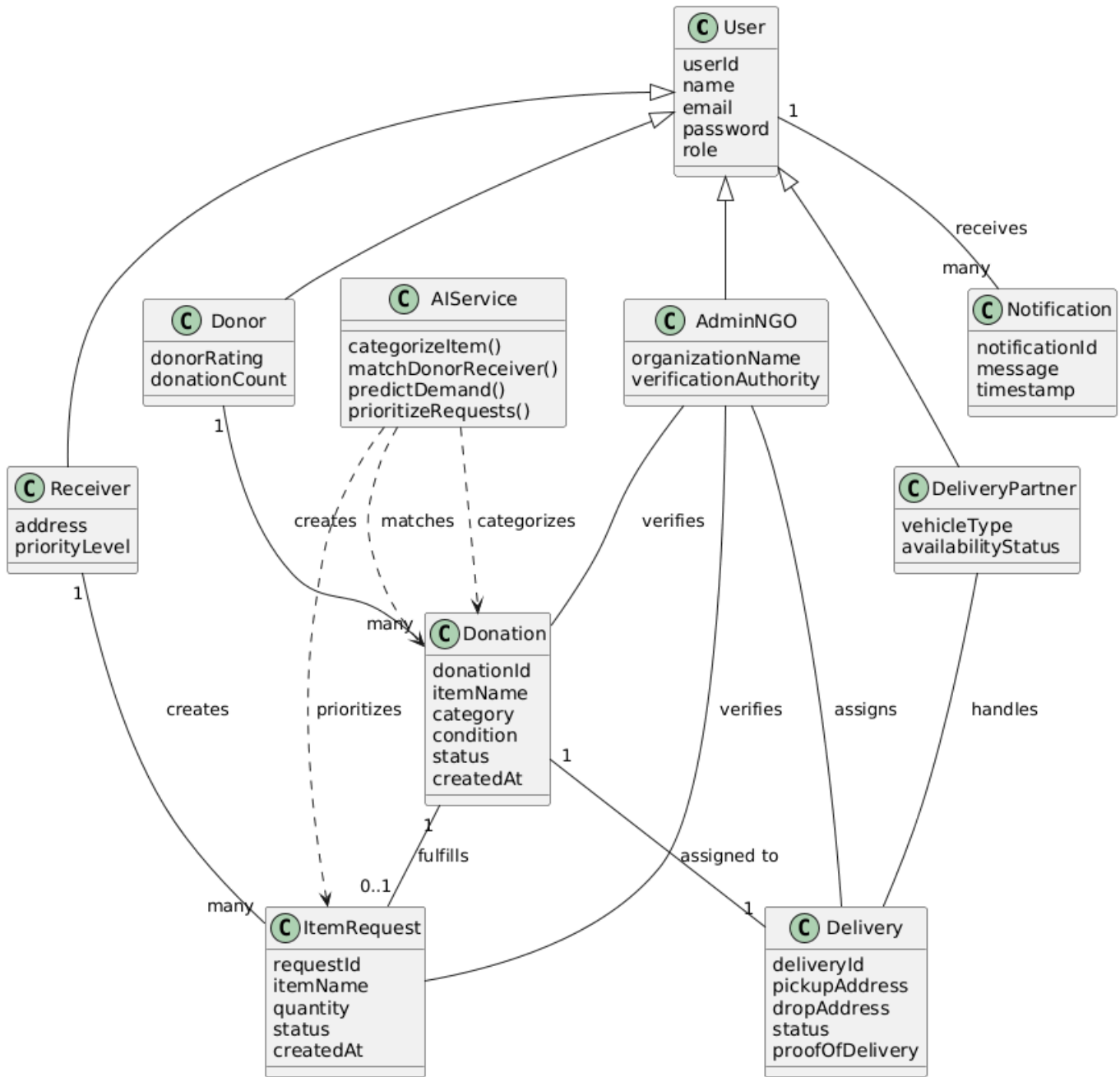
1.3.3 ADMIN



1.3.4 DELIVERY PARTNER



1.4 CLASS DIAGRAM



2. DATABASE DESIGN

2.1 USER TABLE

Field Name	Data Type	Description
id	INT (PK)	Unique user identifier
name	VARCHAR(100)	Full name of the user
email	VARCHAR(100)	Unique email address
password	VARCHAR(255)	Encrypted password
role	ENUM	donor / receiver / delivery / admin
phone	VARCHAR(15)	Mobile number (linked to Aadhaar OTP)
address	TEXT	Residential address
aadhaar_verified	BOOLEAN	Aadhaar OTP verification status
aadhaar_verified_at	TIMESTAMP	Date & time of OTP verification
verification_method	VARCHAR(30)	OTP
is_active	BOOLEAN	Account activation status
created_at	TIMESTAMP	Account creation timestamp

2.2 DONATION TABLE

Description	Field Name	Data Type
Unique donation ID	id	INT (PK)
Donor user reference	donor_id	INT (FK → users.id)
Donation title	title	VARCHAR(100)
Item description	description	TEXT

Food / Clothes / Books etc	category	VARCHAR(50)
Number of items	quantity	INT
pending / verified / assigned / picked / delivered / rejected	status	ENUM
Donation created time	created_at	TIMESTAMP

2.3 DONATION IMAGES TABLE

Description	Field Name	Data Type
Image ID	id	INT (PK)
Donation reference	donation_id	INT (FK → donations.id)
Image file path	image_url	VARCHAR(255)

2.4 RECEIVER REQUEST TABLE

Description	Field Name	Data Type
Request ID	id	INT (PK)
Receiver reference	receiver_id	INT (FK → users.id)
Requested item category	category	VARCHAR(50)
Request description	description	TEXT
Quantity required	quantity	INT
pending / approved / matched / rejected / fulfilled	status	ENUM
Request date	created_at	TIMESTAMP

2.5 VERIFICATION LOG

Description	Field Name	Data Type
Verification ID	id	INT (PK)
Donation reference	donation_id	INT (FK → donations.id)
Request reference	request_id	INT (FK → requests.id)
Verifier (NGO/Volunteer)	volunteer_id	INT (FK → users.id)
approved / rejected	action	ENUM
Verification comments	remarks	TEXT
Verification time	verified_at	TIMESTAMP

2.6 INVENTORY TABLE

Description	Field Name	Data Type
Inventory ID	id	INT (PK)
Donation reference	donation_id	INT (FK → donations.id)
Item category	category	VARCHAR(50)
Available quantity	quantity	INT
Stock availability	available	BOOLEAN

2.7 DELIVERY TABLE

Description	Field Name	Data Type
Delivery ID	id	INT (PK)
Donation reference	donation_id	INT (FK → donations.id)

Request reference	request_id	INT (FK → requests.id)
Delivery user	delivery_partner_id	INT (FK → users.id)
Donor pickup location	pickup_address	TEXT
Receiver drop location	drop_address	TEXT
assigned / picked / in_transit / delivered	status	ENUM
Delivery proof	proof_image	VARCHAR(255)
Delivery completion time	delivered_at	TIMESTAMP

2.8 NOTIFICATION

Description	Field Name	Data Type
Notification ID	id	INT (PK)
Target user	user_id	INT (FK → users.id)
Notification content	message	TEXT
Read status	is_read	BOOLEAN
Notification time	created_at	TIMESTAMP

3. TEST CASES

3.1 REGISTRATION MODULE

Test Case ID	Test Scenario	Input Data	Expected Result
TC_REG_01	Register with valid details	Name, Email, Phone, Password, Role	User registered successfully
TC_REG_02	Register with existing email	Duplicate email	Error message displayed
TC_REG_03	Register without mandatory fields	Empty name/email	Validation error shown
TC_REG_04	Register with invalid email format	abc@	Invalid email error
TC_REG_05	Register and complete Aadhaar OTP verification	Valid OTP	Aadhaar verified successfully
TC_REG_06	Register with invalid OTP	Wrong OTP	OTP verification failed

3.2 LOGIN MODULE – TEST CASES

Test Case ID	Test Scenario	Input Data	Expected Result
TC_LOG_01	Login with valid credentials	Email + Password	Login successful
TC_LOG_02	Login with invalid password	Wrong password	Error message displayed
TC_LOG_03	Login with unregistered email	Unknown email	User not found

TC_LOG_04	Login without Aadhaar verification	Valid credentials	Access denied
TC_LOG_05	Role-based redirection	Admin / Donor / Receiver	Redirect to correct dashboard

3.3 DONOR MODULE

Test Case ID	Test Scenario	Input Data	Expected Result
TC_DON_01	Create donation with valid details	Item, category, quantity	Donation created
TC_DON_02	Upload donation images	Valid image files	Images uploaded
TC_DON_03	Submit donation for verification	Donation data	Status set to "Pending"
TC_DON_04	Edit donation before verification	Updated details	Donation updated
TC_DON_05	Delete donation before verification	Donation ID	Donation deleted
TC_DON_06	Track donation status	Donation ID	Status displayed correctly
TC_DON_07	View donation history	Donor account	Donation list shown

3.4 RECEIVER MODULE

Test Case ID	Test Scenario	Input Data	Expected Result
TC_REC_01	Create request with valid data	Item, quantity	Request created
TC_REC_02	Submit request for approval	Request data	Status set to "Pending"
TC_REC_03	Request approval by NGO	Approved request	Status updated

TC_REC_04	Request rejection	Invalid request	Rejection reason shown
TC_REC_05	Track request status	Request ID	Correct status shown
TC_REC_06	Receive donation	Delivery confirmation	Request marked fulfilled
TC_REC_07	View request history	Receiver account	History displayed

3.5 ADMIN MODULE

Test Case ID	Test Scenario	Input Data	Expected Result
TC_ADM_01	Login as admin	Admin credentials	Admin dashboard loaded
TC_ADM_02	View all users	User list	Users displayed
TC_ADM_03	Manage user roles	Role update	Role updated successfully
TC_ADM_04	View donations & requests	System data	Data displayed correctly
TC_ADM_05	Monitor deliveries	Delivery data	Delivery status shown
TC_ADM_06	Generate reports	Date range	Report generated
TC_ADM_07	Resolve dispute	Dispute case	Issue resolved